

Thomas Hayden

Backend Software Engineer

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<https://t0ckendl0l.github.io/>

PROFESSIONAL SUMMARY

Backend Software Engineer specializing in Python-based API development and cloud-deployed services. Experienced in building, deploying, and monitoring production-ready backend applications using FastAPI, SQL, Docker, and AWS. Strong focus on reliability, performance, security, and clean service architecture, with proven ownership in high-availability environments.

SKILLS

- Languages - Python, SQL, JavaScript (basic), Bash
- Cloud & DevOps - AWS (EC2, S3, IAM, CloudWatch)
- Backend & APIs - FastAPI, Flask, REST APIs, JSON, OpenAPI, Authentication
- Docker, Docker Compose, Git, GitHub, AWS Well-Architected Framework
- Databases - SQLite, PostgreSQL, SQLAlchemy
- Data & Visualization - Matplotlib, Bokeh, Seaborn

EXPERIENCE

PRODUCTION-GRADE GPT CHATBOT API

November 2025 - December 2025

Python, FastAPI, AWS, Docker, Nipomo CA

- Designed and implemented a stateful chatbot API supporting multi-turn conversations with tool usage.
- Deployed containerized backend services on AWS EC2 using Docker for reproducible environments.
- Configured IAM roles and policies following least-privilege access principles.
- Incorporated AWS CloudWatch for logging, monitoring, and application health visibility.
- Implemented request validation with Pydantic and FastAPI for reliability and safety.
- Reduced average response latency to approximately 1.2 seconds while maintaining low per-request cost.
- Applied AWS shared responsibility and Well-Architected Framework principles in deployment decisions.

AI-DRIVEN SMART CITY TRAFFIC & ENERGY SIMULATION

October 2025 - November 2025

Full Sail University, Nipomo CA

- Developed a simulation environment modeling traffic flow and energy usage in a smart-city context.
- Implemented reinforcement learning agents to optimize congestion reduction and energy efficiency.
- Designed decision logic for rerouting, waiting, and adaptive control strategies.
- Collected, analyzed, and visualized performance metrics to evaluate agent behavior and system impact.

REAL-TIME FACE AGE PREDICTION SYSTEM

September 2025 - October 2025

Full Sail University, Nipomo CA

- Built a real-time webcam application for face detection and age classification.
- Integrated pretrained deep learning models using OpenCV DNN pipelines.
- Implemented live bounding boxes, age labels, and screenshot capture functionality.
- Optimized inference performance to support real-time execution on consumer hardware.

EDUCATION

Master of Science in Computer Science (M.S.)

August 2026

Full Sail University, Winter Park, FL

- Focus: Artificial Intelligence, Software Engineering.

Bachelor of Science in Game Art

May 2025

Full Sail University, Winter Park, FL

- Focus: Environmental Art.